

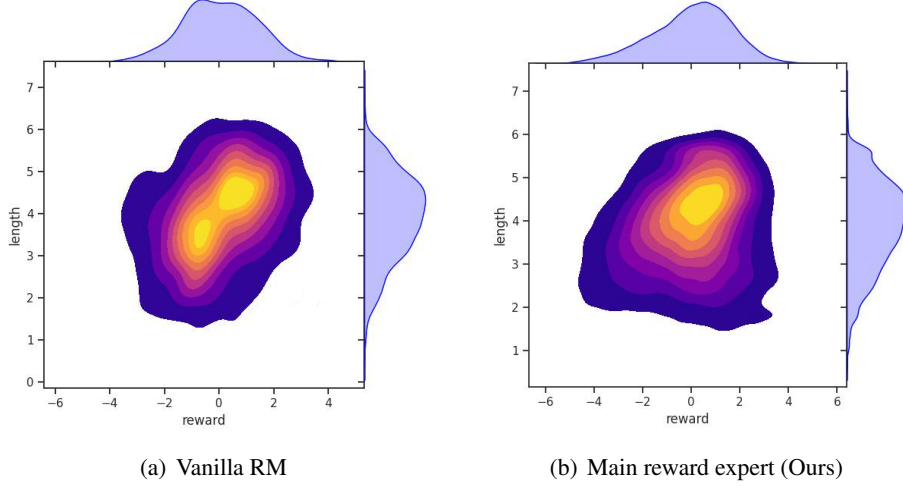


Figure 2: Distributions of Reward vs. log-scaled Length in vanilla RM and our proposed method. The results illustrate that Vanilla RM tends to assign higher rewards to longer responses, whereas our proposed method achieves a more uniform distribution of rewards across different response lengths.

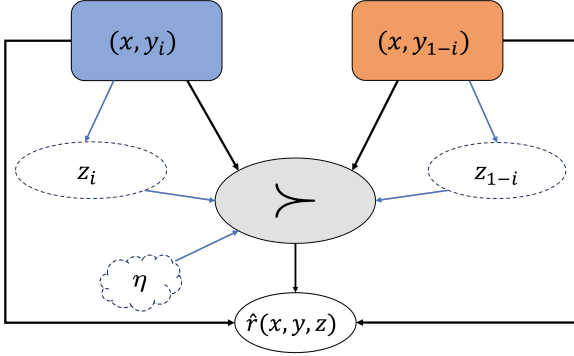


Figure 3: Causal structure of preference-based reward modeling. Length bias, represented as a confounding factor z , influences the model’s learning of true human preferences (x, y) .

z might be correlated with (x, y) , potentially due to their sharing of the same causal parent or there being biases during data collection, like annotators are inclined to favor longer sentences, which is expected.

In the presence of confounding effects, the goal of preference-based reward modeling is to learn a biased reward function $\hat{r}(x, y, z)$ that best matches the stated preferences. There are state features (x, y) that are causal with respect to the true human intent but are unobserved by the learning agent and considered non-robust features effortlessly learned. In our case, we suppose that length bias is one vital element strongly related to (x, y) . Actually, this situation can hardly be completely avoided, and our motivation is to reduce the probability of the path directed by the blue

arrow. Furthermore, unobserved human bias and noise induced by annotators, such as inappropriate judgments on certain individuals, groups, races, sexes, etc., denoted by η , also affect the preference labels.

As a consequence, the learned reward $\hat{r}$ has risk to achieve actually low yet even near-perfect performance on a held-out test dataset with following two factors: 1) causal factor and nuisance factor can be correlated by the con-founder factor (x, y) , 2) RM easily find a shortcut exists between causal factor and reward function in the distribution of test set and can not be extrapolated to out-of-domain dataset. This might be reflected when this misspecified reward model $\hat{r}(x, y, z)$ guide the LM agent to align with human intention using PPO, as a result of distribution drift.

4.3 Challenges

The spurious correlation between reward scalars and response lengths can be problematic, as it may result in biased or sub-optimal model behavior. However, due to the unobserved nature of this spurious attribute when performing a traditional ERM-based training paradigm, it can be challenging to investigate and address during the reward modeling stage. Inspired by the debiasing framework of ensemble model methods, we posit that length bias can be alleviated through a robust learning approach, which involves disentangling the underlying features and feeding them into distinct experts. This assumption is grounded

in the notion that the bias in the length of data can be attributed to the confounding effect of certain features, which can be disentangled through representation learning. Based on the aforementioned observations and insights, in the next section, we will propose a simple and effective method to mitigate the length bias during the reward modeling stage.

5 Proposed Method

The section describes the framework and algorithm of our method. We introduce an approach to establish a debias framework that can significantly mitigate the length bias at the RLHF stage.

5.1 PoE Framework

In this study, our framework is mainly built based on the procedure of the reward modeling phase, as illustrated in Figure 4. To learn a reward model from preferences, we assume access to a set of pairwise preference pairs. Specifically, after a batch of N data consisting of an equal quantity of positive and negative sentences passes through our framework, the learning process begins.

Products-of-Experts. In our study, we explore a reward modeling method that utilizes different experts to separately learn the true human intent and the length bias. We employ the Products-of-Experts (Hinton, 2002) technique to train a robust reward model. Specifically, our ensemble method can be formulated as:

$$\hat{r}(x, y) = \text{Softmax}(\log(r_\phi(x, y)) + \log(r_\psi(x, y))).$$

Equivalently, $\hat{r}(x, y) \propto r_\phi(x, y) \circ r_\psi(x, y)$, where $r_\psi(x, y)$ is the output of the bias-only model and $r_\phi(x, y)$ is that of the main reward model.

To ensure the main reward expert and bias-only reward expert learn different content, we applied constraints based on empirical observations. The main reward model utilized a larger language expert (e.g., 7B LLaMA (Touvron et al., 2023)) and a normal learning rate to capture human intent. In contrast, the bias-focused expert employed a smaller model (e.g., 560M BLOOMZ (Muennighoff et al., 2022)) and a higher learning rate, typically three times that of the main expert. Previous studies (Mhaskar et al., 2016; Wilson et al., 2018) showed that smaller models with larger learning rates tend to learn simpler and coarser information. This differentiation in model size and

learning rate aims to balance comprehensive understanding with the identification and mitigation of potential biases (Geirhos et al., 2020).

5.2 Injecting Noise into Bias-only Expert

In order to ensure that the bias-only model captures the length bias present in the input, we employ a technique of injecting random noise into the input. This intentional introduction of noise serves the purpose of disrupting the semantic information within the input, thereby aiding the model in effectively learning the length bias. Thus, the perturbed inputs can be expressed as $X' = X + N$, where X' represents the new, noisy input, and N denotes the Gaussian noise added to the token embeddings. By facilitating the collaboration between bias-only experts and main experts in modeling human preference, we enable the bias-only experts to effectively capture length bias while preventing the main model from learning length bias.

5.3 Training & Inference

During the training phase, the main expert and the bias-only expert are jointly optimized by maximizing the following likelihood function to optimize the reward function:

$$-\mathbb{E}_{(x,y) \sim \mathcal{D}}[\log(\sigma(\hat{r}(x, y_i) - \hat{r}(x, y_{1-i})))] \quad (2)$$

The main expert is initialized based on an SFT model, while the bias-only expert is initialized using a pretrained model. Both reward models add a linear layer on top of the final Transformer layer, which generates the final scalar prediction for the reward signal.

During the PPO stage, we exclusively rely on the main expert to provide rewards, while discarding the bias-only expert. Since the bias-only expert is typically smaller in size, our approach does not significantly increase computational overhead.

6 Experiments

6.1 Settings

Datasets We utilize the Helpful and Harmless (HH) dataset (Bai et al., 2022b) from Anthropic as our experimental dataset and rm-static² for training our reward model and for participation in PPO. The HH dataset provides a response

²<https://huggingface.co/datasets/Dahoas/rm-static>

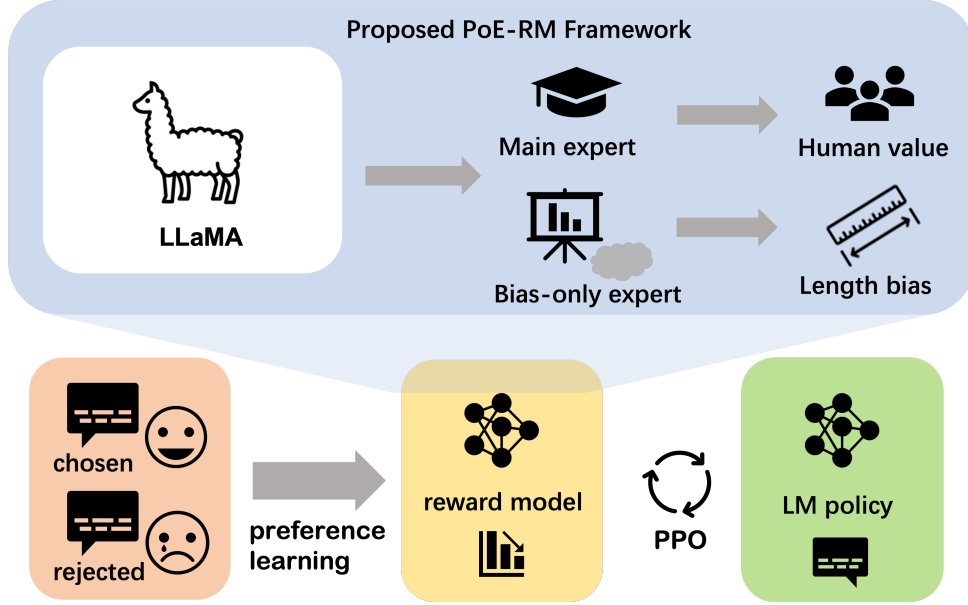


Figure 4: Pipeline of the proposed method. The proposed PoE-based reward modeling approach consists of two experts: the main expert, which learns the true human value, and the bias-only expert, which focuses on capturing the length bias.

and a rejected response for each query based on human preferences, specifically focusing on responses that are helpful and harmless. In addition to that, our SFT data incorporated the 52k instruction dataset constructed by Alpaca³ and the ChatAlpaca⁴ dataset containing multi-turn dialogues.

Models In our experimental setup, we primarily build upon LLaMA and BLOOMZ models, utilizing models with a parameter size of 7B. Inspired by the work of (Ouyang et al., 2022), who employ SFT models as initial models for PPO, we perform SFT on the Alpaca and ChatAlpaca datasets.

SFT Hyper-parameters During the SFT phase, we utilize a learning rate of $3e^{-5}$ and train for three epochs without early stopping. We employ a warmup period of 0.3 epochs, followed by a linear decay to 0. The fine-tuning process was conducted on a device with eight Nvidia A100 GPUs. Each GPU handled four queries, resulting in a batch size of 32. Responses are truncated to 512 tokens, while the total length of both queries and responses was truncated to 2048 tokens. We incorporate specific prompts, such as "Human:" or "Assistant:", during the concatenation of input queries and output responses. These prompts are added to provide

context and distinguish between human-generated and assistant-generated responses.

RLHF Hyper-parameters During the reward modeling training phase, our main expert and policy model remained consistent. The learning rate for both is set to $5e^{-6}$. As for the bias-only expert, we utilize a smaller model, the 560m Bloomz, with a fixed learning rate of $8e^{-6}$.

In the PPO framework, we perform reward score normalization and clipping, with a clip value of 0.8. We employ the clipped surrogate objective of PPO for optimization. The token-level KL penalty coefficient β is set to 0.05. For each query, we collect 4 roll-out samples using nucleus sampling. The sampling temperature is set to 0.8, top-p is set to 0.9, repetition penalty is set to 1.1, and the maximum output token length is set to 512. The policy model has a learning rate of $9e^{-7}$, while the value model utilize a learning rate of $2e^{-6}$. These specific training details are implemented to optimize the performance and convergence of the models during the training process.

Baselines In this study, we propose a method primarily aimed at mitigating length bias in the reward model. Therefore, our baselines include the SFT model, the PPO model trained with the vanilla reward model, and the PPO model trained exclusively with the bias-only reward expert.

³https://github.com/tatsu-lab/stanford_alpaca

⁴<https://github.com/cascip/ChatAlpaca>